Human-Centered Data Collection for Machine Learning

Lessons from Survey Research

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May 12, 2025

AAPOR Conference Short Course

Four hours, one argument

1. Introduction: Why data quality matters
2. Survey data collection basics
3. Training data collection
4. **Measurement:** Are the labels correct?
5. **Representation:** Who labels?
6. **Sampling:** Which instances to label?
7. Ethics of data work
8. Practical advice
9. Discussion

We will take breaks. Ask questions anytime.

Who we are

Stephanie Eckman – University of Maryland

Survey methodologist studying how data collection design affects data quality – now applied to ML training data.

Frauke Kreuter – LMU Munich / University of Maryland

Survey methodology, data science, and the intersection of social science methods with machine learning.

ML models are only as good as their training data

"Everyone wants to do the model work, not the data work."

— Sambasivan et al. (2021)

Data quality problems are the most common cause of ML system failures in production.

(Sculley et al., 2015)

The shift to data-centric AI

Traditional ML: fix the data, iterate on the **model**.

Data-centric AI: fix the model, iterate on the **data**.

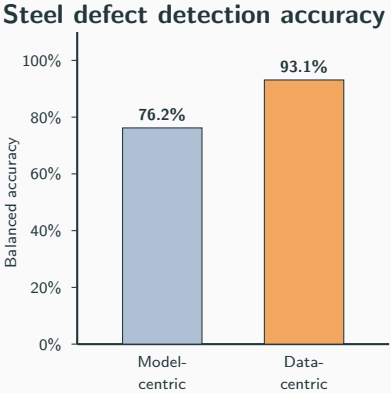
Survey researchers have been iterating on data quality for **decades**. We have tools for this.

(Ng, 2021)

Better data beats a better model

"A reasonable algorithm with good data is preferable to a great algorithm with not-so-good data."

— Andrew Ng, 2021



Improving the **data** yielded a 17 percentage point gain.
Improving the model yielded almost none.

(Ng, 2021)

Industry leaders worry about data quality

> *"The bias I am most nervous about is the bias of the human feedback raters."*

— Sam Altman, March 2023

The people who label training data shape the model.
Their biases become the model's biases.

This is a **data collection** problem —
exactly the kind survey researchers know how to study.

Survey researchers bring a unique perspective

- Frameworks for thinking about data quality (Total Survey Error)
- Experience designing instruments that minimize bias
- Understanding of how question design affects responses
- Methods for assessing and reducing measurement error
- Decades of research on who responds and why it matters

Survey Data Collection Basics

The Total Survey Error framework

Two families of error:

Measurement

Is the answer correct?

Validity, measurement error, processing error

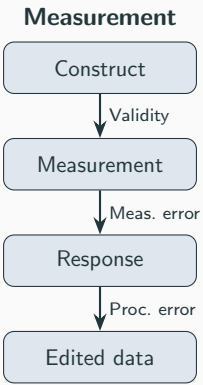
Representation

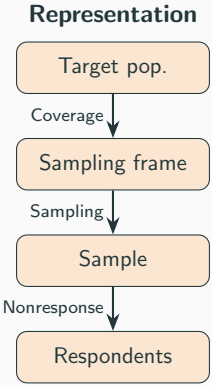
Who answered?

Coverage, sampling, nonresponse, adjustment

This framework organizes our entire course.

Total Survey Error: the full picture





Measurement: three sources of error

Validity Does the question measure what we intend?

Measurement error Systematic or random deviations from the true value

Processing error Errors introduced during coding, editing, or data entry

Each of these has a direct analogue in training data collection.

Representation: three sources of error

Coverage Who is missing from the frame?

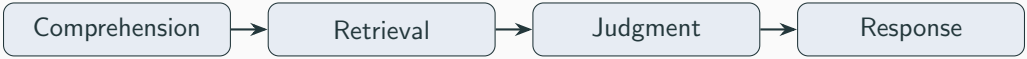
Sampling Which units are selected?

Nonresponse Who refuses or is unreachable?

Each of these has a direct analogue in training data collection.

How survey respondents answer questions

The cognitive model of survey response:



Errors can enter at **every stage**.

The same model applies to annotators labeling data.

Training Data Collection

ML models need humans to learn from

Supervised learning requires labeled examples:

- A model sees an image $\rightarrow$ a human says "cat"
- A model reads a tweet $\rightarrow$ a human says "hate speech"
- A model generates text $\rightarrow$ a human says "response A is better"

Without these human judgments, the model cannot learn what "correct" means.

The quality of these labels **directly shapes** model behavior.

Where does training data come from?

1. **Found data** – web scraping, social media, government records
2. **Elicited data** – surveys, experiments
3. **Human-labeled data** – human annotators assign labels to items

This course focuses on **human-labeled data**:
the labels that teach ML models what is "correct."

Two kinds of human labeling

Training data collection

Tend to be **objective** tasks

- Image classification (dog/cat/other)
- Named entity recognition
- Object detection in images
- Document categorization

Labelers: crowdworkers, experts, researchers, graduate students

Human feedback collection

Tend to be **subjective** tasks

- Which response is "better"?
- Is this content harmful?
- Rate quality on a 1–7 scale
- Is this summary accurate?

Used to fine-tune LLMs via RLHF (reinforcement learning from human feedback)

Example: image classification

A labeler sees an image and selects a category:

What kind of animal is this?

- ○ Bird
- • Not_Bird
- ○ Unable_to_tell

SUBMIT

(NLPerspectives Workshop, 2024)

Seems simple, but consider:

- What counts as a "bird"?
- A toy bird? A painting?
- What if you can't tell from the image?

Even "objective" tasks require interpretation.

Example: RLHF rating interface

The interface used to train InstructGPT / ChatGPT:

Instruction: Summarize the following news article.

[Article text here]

Output A: Article summary **Rating (1=worst, 7=best):** 1 2 3 4 5 6 7

Does it contain sexual content? ◯ Yes ◯ No

Does it contain violent content? ◯ Yes ◯ No

Does it give harmful advice? ◯ Yes ◯ No

(Ouyang et al., 2022)

Example: named entity recognition

Annotators highlight text spans and assign entity types:

East Repair Inc.	1912 Harvest Lane	11/02/2019
SELLER	ADDRESS	DATE	AMOUNT

The labeler decides: Where does an entity start and end?
What type is it? These are **judgment calls**.

Who labels training data?

Crowdworkers – Amazon MTurk, Appen, Scale AI, Surge AI

Fast, cheap, large scale. Variable quality and training.

Domain experts – doctors, lawyers, linguists

High quality, expensive, scarce. Used for specialized tasks.

Researchers / graduate students – small scale

Often for academic datasets. Potential bias toward researcher's hypotheses.

LLMs – GPT-4, Claude, etc.

Increasingly common. Fast, but carries systematic biases.

These annotation tasks are a lot like surveys

Annotation interface

Does this tweet contain hate speech?

- ◯ Yes
- ◯ No

Does this tweet contain offensive language?

- ◯ Yes
- ◯ No

Survey questionnaire

How safe is your neighborhood?

- ◯ Very safe
- ◯ Somewhat safe
- ◯ Not safe

How confident are you in this answer?

◯ Very ◯ Somewhat ◯ Not

Same structure: radio buttons, response scales, skip logic.

Same design → same potential for error.

Labelers are people, not machines

Behind every training dataset are **human decisions**.

- Annotators interpret guidelines
- They bring their own backgrounds and biases
- They experience fatigue, boredom, and confusion
- They are affected by task design, pay, and working conditions

Sound familiar? These are the same issues survey researchers study in **interviewers** and **respondents**.

Total Survey Error maps onto training data

TSE Concept	Survey	Training Data
Measurement	Question $\rightarrow$ answer	Task $\rightarrow$ label
Validity	Q. measures construct?	Label reflects concept?
Meas. error	Interviewer effects	Annotator effects
Representation	Who responds?	Who labels?
Coverage	Frame misses units	Pool misses instances
Sampling	Which units selected?	Which instances labeled?
Nonresponse	Who refuses?	Who drops out/skips?

Two questions organize the rest of this course

Measurement: Are the labels correct?

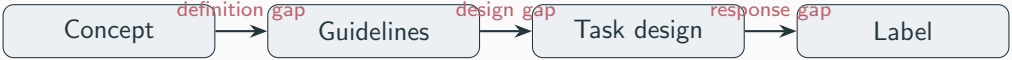
Representation: Who labels?

The same two families of error from Total Survey Error.

We will tackle each one in turn.

Measurement: Are the Labels Correct?

The construct-to-label path is long



At each step, error can enter.

Survey researchers know this path well.

Annotation design affects the labels you get

Kern et al. (2023) ran an experiment:

Same tweets, different annotation designs → **different labels**.

Design choices that mattered:

- Number of response categories
- Whether "offensive" was a separate category
- Task framing and instructions

The label is not a property of the text.
It is a product of the **measurement instrument**.

Hate speech / offensive language classification

Different instruments, different "ground truth"

Kern et al. (2023) tested multiple annotation designs for the **same** hate speech classification task.

Key finding: Models trained on labels from different designs had **different accuracy levels** and **different error patterns**.

Implication: The "ground truth" depends on how you ask.

Pre-labeling can introduce automation bias

Beck et al. (2025) studied what happens when annotators see **model-suggested labels** before making their own judgment.

- Annotators anchor on the pre-label
- They are less likely to disagree with the suggestion
- This is **automation bias** – the same phenomenon documented in aviation, medicine, and survey editing

Pre-labeling saves time but may **reduce label independence**.

The order of items affects labels

Beck et al. (2024) found that the **sequence** in which annotators see items affects their labeling decisions.

Survey researchers call these **context effects**:

- Assimilation: current judgment pulled toward previous item
- Contrast: current judgment pushed away from previous item
- Fatigue: quality degrades over the session

Implication: Randomize item order across annotators.

Lessons from 80 years of question design research

Survey research offers tested principles:

- Avoid ambiguous terms (define "hate speech")
- Provide balanced response options
- Use consistent scales across items
- Avoid double-barreled questions
- Pretest the instrument before full deployment

(Krosnick and Presser, 2010; Sudman and Bradburn, 1982)

Discussion

Discussion: Measurement

1. Think of a labeling task in your work.
   What design choices could affect the labels?

2. Have you encountered situations where the "ground truth" depended on how the question was asked?

Representation: Who Labels?

Who provides the labels?

Domain experts Doctors, lawyers, linguists – high quality, expensive, scarce

Trained annotators Hired and trained for the task – moderate cost

Crowdworkers MTurk, Prolific, Scale AI – cheap, fast, variable quality

The researchers themselves Small scale, potential bias

LLMs Increasingly used, but with systematic biases

Each choice shapes the labels – and the resulting model.

Nonresponse bias applies to labeling too

In surveys, nonresponse bias arises when:
people who respond differ from people who don't.

In labeling:

- Which annotators accept the task? (self-selection)
- Which annotators complete it? (attrition)
- Which annotators pass quality checks? (filtering)

At each stage, the remaining annotators
may **differ systematically** from those who left.

Labeler characteristics affect the labels

Annotator demographics predict label distributions.

Otterbacher et al. (2018): Annotator gender and age affected how they labeled stereotype content in image search.

Al Kuwatly et al. (2020): Annotator demographics predicted systematic differences in toxicity labels.

This is not noise – it is **systematic measurement error** correlated with annotator characteristics.

Labeler diversity affects model behavior

If your annotators are **demographically homogeneous**:

- Labels reflect one group's perspective
- Models learn that perspective as "ground truth"
- Performance degrades for underrepresented groups

Survey researchers understand this:
a sample must **represent** the target population.

But who is the "target population" for annotators?

Can LLMs replace human labelers?

Increasingly, teams use GPT-4 or similar models to generate labels.

Risks:

- LLMs reproduce biases in their training data
- **Model collapse:** models trained on model-generated data lose diversity over generations (Shumailov et al., 2023)
- LLM labels are not independent of each other
- We lose the **signal in human disagreement**

LLM labels may be useful, but they are not a free replacement for thoughtful human annotation.

Discussion

Discussion: Representation

1. Who labels in your organization?
   How were they recruited and selected?

2. If labeler characteristics affect labels,
   what does "representative" mean for an annotator pool?

Sampling: Which Instances to Label?

You can't label everything

Labeling is expensive. You must choose which instances to label.

Options:

- Simple random sampling from the pool
- Uncertainty sampling (where the model is confused)
- Diversity sampling (cover the feature space)
- Active learning (iterative, model-guided selection)

(Monarch, 2021)

How many labels do you need?

In surveys: sample size is determined **before** data collection
(power analysis, margin of error, design effect).

In active learning: sample size is determined **during** collection
(stop when accuracy plateaus or budget runs out).

There is no universal formula.
It depends on task complexity, class balance, and model architecture.

Sampling for Labels
vs. Sampling for Surveys

You know sampling – but ML uses these words differently

You have spent careers thinking about:
probability sampling, stratification, clustering, sample size.

The active learning literature uses **the same vocabulary**
but **shifts the meanings**.

The next several slides map the terminology precisely.
The collisions are not minor – they are fundamental.

Monarch's four types of "diversity sampling"

Monarch (2021) uses **diversity sampling** as an umbrella for all non-uncertainty-based sampling:

1. **Model-based outliers** – items far from training data in feature space
2. **Cluster-based sampling** – k-means on features, sample from each cluster
3. **Representative sampling** – sample items that look like the unlabeled pool
4. **Real-world diversity** – ensure demographic groups are covered

He notes: "diversity sampling goes by different names in different fields, such as representative sampling, stratified sampling."

The same words mean different things

Term	Survey Sampling	Active Learning
Stratification	Non-overlapping subgroups of the *population*; sample from each stratum	Bin items by *model confidence* or *predicted label*; sample from each bin
Clustering	Select *groups of units* (schools, blocks) as PSUs for cost efficiency	Group *data points* by similarity in feature space; sample near centroids
Representative	Sample mirrors the *population* via probability sampling + weighting	Sample mirrors the *unlabeled pool* in feature space
Diversity	Via stratification; ensure population subgroups represented	Umbrella term for all non-uncertainty sampling

“Stratified sampling” means something different in ML

In surveys:
Define non-overlapping subgroups of a *population* **before** data collection. Sample from each stratum with known inclusion probabilities.

In active learning:
Bin unlabeled items by *model confidence* or *predicted label*. Sample equal amounts from each bin to avoid oversampling easy, high-confidence cases.

Strata in AL are defined by **model output**, not population characteristics. They have no inclusion probabilities.
And they **change** every time the model retrains.

Example: Monarch's "stratify by confidence"

Model confidence intervals

0–10% | 10–20% | 20–30% | 30–40% | 40–50% | 50–60% | 60–70% | 70–80% | 80–90% | 90–100%

Sample **equal** n from each bin

Without this, you oversample easy cases

A survey statistician would call this **post-stratification** on a model-generated variable – not stratified sampling.

(Monarch, 2021)

“Clustering” means something different in ML

In surveys:
Select *groups of operational units* (schools, blocks, households) as primary sampling units for **cost efficiency**.
Then survey units within selected clusters.

In active learning:
Run k-means or similar on *feature-space embeddings*.
Sample items nearest each centroid to maximize **coverage of the feature space**.

Survey clusters are defined by **geography and logistics**.
AL clusters are defined by **similarity in a learned representation**.
Same word, completely different unit of analysis and purpose.

“Representative” means something different in ML

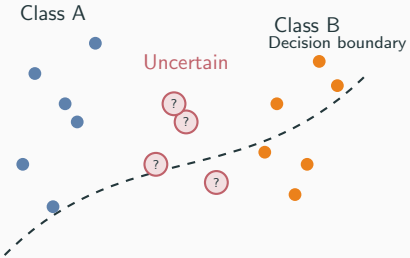
In surveys:
A sample is representative when it mirrors the *population* via probability sampling and appropriate weighting.

In active learning:
A sample is “representative” when it mirrors the *unlabeled data pool* in feature space.

Key difference:
AL has no concept of inclusion probabilities or design weights.
There is no Horvitz-Thompson estimator.
“Representative” is informal, not inferential.

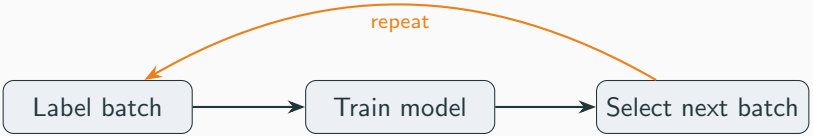
Uncertainty sampling has no survey analogue

Select instances where the **model is most uncertain** (near the decision boundary, roughly 50/50 predictions).



This requires a **trained model** – surveys don't have one.
It is fundamentally model-driven, not design-driven.

Active learning is iterative; most surveys are not



Closest survey analogue: **adaptive/responsive survey designs** (changing protocol based on accumulating data).

Key difference: AL optimizes **model accuracy**.
Surveys optimize **population estimates**.

Sample size: formula vs. learning curve

Surveys

n determined **before** collection:

- Power analysis
- Margin of error
- Design effect

Clear stopping rule.

Active learning

n determined **during** collection:

- Monitor learning curve
- Stop when accuracy plateaus
- Or when budget runs out

No universal formula.

What survey statisticians can bring to ML

- **Probability-based inference** – AL has no guarantee that the labeled sample represents any population
- **Design-based estimation** – Horvitz-Thompson, inclusion probabilities, design weights
- **Rigorous uncertainty quantification** – confidence intervals tied to the sampling design
- **Decades of research** on nonresponse, coverage, and measurement error

The ML field could benefit enormously from this expertise.

What ML can bring to survey statisticians

- **Adaptive, model-driven** sampling strategies
- Optimizing for **prediction** rather than just estimation
- **Active sampling for finite population inference**
  – an emerging bridge between the fields
- Efficient use of limited labeling budgets

The best path forward combines both traditions.

Discussion

Discussion: Sampling

1. Where else do these fields talk past each other using the same vocabulary?

2. In your work, when would you choose survey-style probability sampling over active learning – or vice versa?

Ethics of Data Work

The people who label your data

Training data is produced by **people working for pay**.

Ethical considerations:

- **Wages:** Many crowdworkers earn below minimum wage
- **Benefits:** No health insurance, no job security
- **Harmful content:** Labelers exposed to violence, hate speech, abuse
- **Global dynamics:** Wealthy countries commission work; lower-income countries perform it

Labelers are exposed to harmful content

Content moderation labelers routinely view:
violence, child abuse, self-harm, hate speech.

- Psychological harm is well-documented
- Workers often lack mental health support
- High turnover signals the toll this takes

If your model requires labels on harmful content,
you have an obligation to protect labelers.

Are labelers human subjects?

Labeling tasks involve human judgment.
Should they require IRB/ethics board review?

- Labelers are performing a task, not being studied – usually
- But research *on* labelers (behavior, accuracy, bias) clearly involves human subjects
- The line is blurry; current regulations lag behind practice

Best practice: Treat labelers with the same ethical care you would give survey respondents.

Discussion

Discussion: Ethics

1. What ethical obligations do ML teams have to the people who label their data?

2. How should labeling tasks involving harmful content be designed to protect workers?

Practical Advice

Concrete recommendations for better training data

1. **Treat labeling as measurement** – apply instrument design principles
2. **Pretest your task** – pilot with a few annotators first
3. **Document your design** – task instructions, response options, labeler selection
4. **Randomize** item order across annotators
5. **Use multiple annotators** – disagreement is signal, not noise
6. **Collect paradata** – time per item, revision patterns, confidence
7. **Report who labeled** – demographics, expertise, selection method

Label collection tools

Open source: Label Studio, Doccano, Prodigy

Commercial: Scale AI, Labelbox, Amazon SageMaker Ground Truth

Crowd platforms: Amazon MTurk, Prolific, Appen, Surge AI

Gaps: Most tools do not collect paradata
(timing, revisions, annotator characteristics)
the way survey data collection platforms do.

This is an opportunity for improvement.

What the ML pipeline is missing from surveys

- **Standardized quality frameworks** (like TSE)
- **Pretesting protocols** for labeling instruments
- **Paradata collection** (timing, process data)
- **Transparent reporting** of data collection methods
- **Labeler selection documentation** (like survey sampling reports)

(Eckman and Kreuter, 2024)

Discussion and Closing

Open Discussion

Questions for discussion

1. What aspects of survey methodology are most transferable to training data collection?
2. What is different enough that new frameworks are needed?
3. How should the survey research community engage with the growing demand for labeled data?

The one thing to remember

Training data collection is data collection.

The same principles that make surveys reliable — careful design, transparent methods, rigorous quality frameworks — apply directly to labeling.

Survey researchers are uniquely qualified to improve this critical piece of the ML pipeline.

Thank you

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Slides and references available at:
`https://github.com/seckman/training_data_course`

Paper: Eckman and Kreuter (2024)

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